

**Standard Operating Procedure (SOP)/Guidelines under Rule 73A of
Rajasthan Minor Mineral Concession Rules, 2017 for
installation/automation of weighbridges to manage and regulate the
transportation of minerals**

1. Details of hardware components:

(a) Mandatory hardware components and its specifications:

S.N.	Item	Quantity	Purpose	Specifications
1	RFID Reader	1	To read the RFID Tag/FastTag pasted on vehicle mirror.	Range: 3000 to 6000 mm Operating frequency: 865-867 MHZ (details Annexure-1)
	For bidirectional vehicle movement, additional RFID on other side of WB having same specification.			
2	Position Sensors	3	To detect and accurately position the vehicle during weighment.	Type: ultrasonic Range: 200 to 2000 mm (details Annexure-1)
3	IP Bullet Camera – C1	1	To capture Top view Photo during the weighment and identification of mineral.	Type: IP Resolution: 2 mega pixel (details Annexure-1)
4	IP Bullet Camera – C2	1	To capture Front view Photo during the weighment and for number plate reading.	Type: IP, lens: Varifocal Resolution: 4 mega pixel (details Annexure-1)
	For bidirectional vehicle movement, additional Camera on other side of WB having same specification.			
5	IP Bullet Camera – C3	1	To capture the image of the digitizer and PC screen during the weighment.	Type: IP Resolution: 2 mega pixel (details Annexure-1)
6	Controller	1	To Manage Weighbridge Hardware and Integrate with Application.	As per Annexure-1
7	Poles	4	For installation of sensors, cameras,	As per Annexure-1
8	Side Railing	-	On both sides, along full length of the weighbridge.	As per Annexure-1
9	Personal Computer	1	To manage the IoT device and Controller using desktop application.	i5 10 th generation or higher
10	PoE network switch	1	To connect all Ethernet devices.	8+2 port (details Annexure-1)

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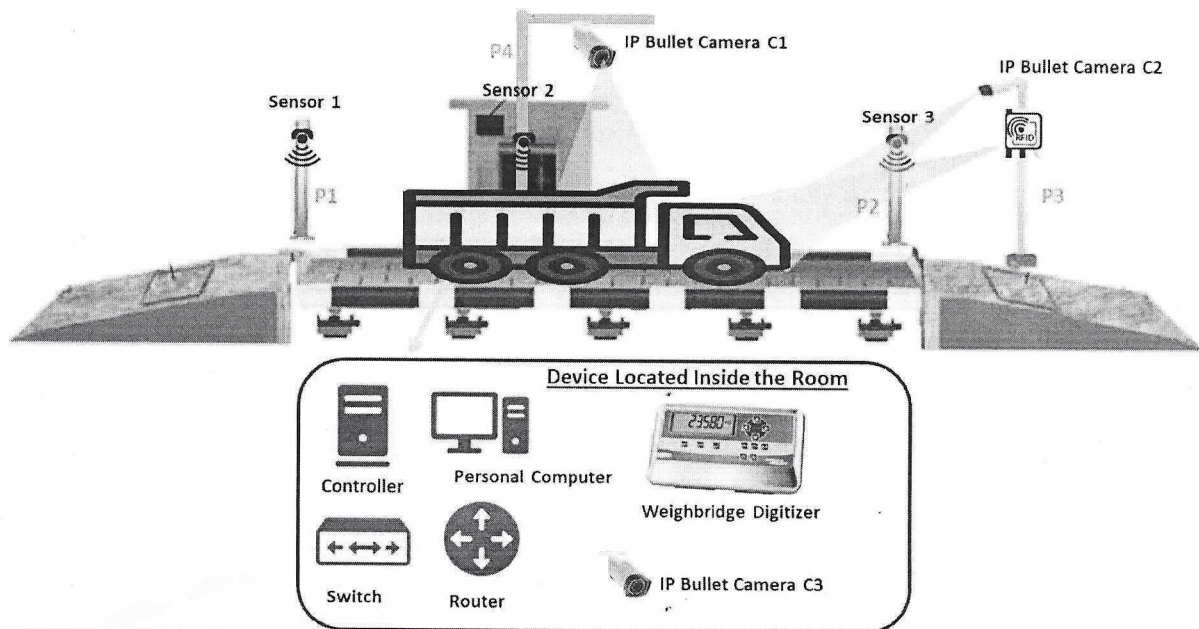
(b) Optional hardware components and its specifications:

S.N.	Item	Quantity	Purpose	Specification
1	LED Display	1	To display various messages /instruction during weighment.	As per Annexure-1
2	PA System/ speaker	1	For Voice based announcement System to guide the driver.	As per Annexure-1
3	Traffic Lights	2	To guide the driver.	As per Annexure-1
4	Boom Barrier with Safety Sensor	1	To allow the vehicle to exit.	As per Annexure-1

(c) Other IT infrastructures and its specifications:

S.N.	Item	Specification
1	Electricity Supply & Power backup	Online backup as per requirement.
2	Internet Connectivity	Dedicated 20+ Mbps Internet Leased Line (ILL) / Broadband / Airfiber connection with; a) Public Static IP; and b) NAT and Port Forwarding support through Service Provider-supported router.

2. Positions for installation of hardware components:

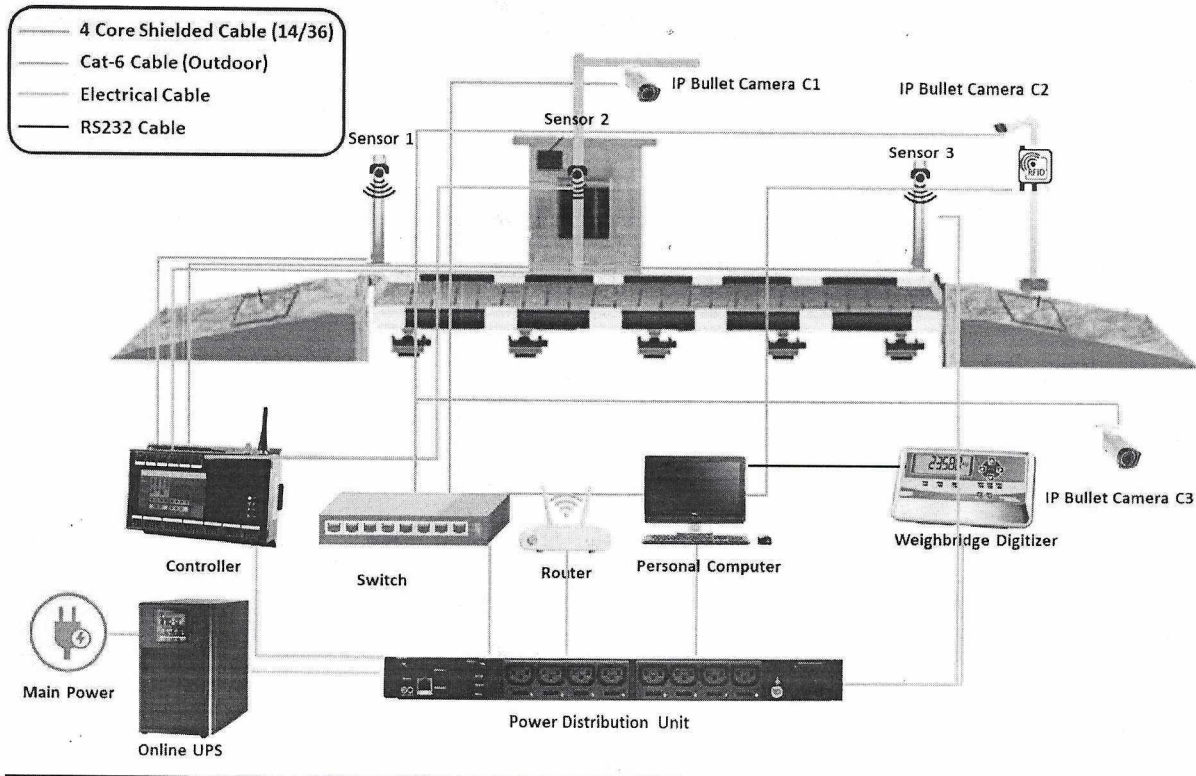


Note: The position of hardware may be changed by ME/AME as per site feasibility

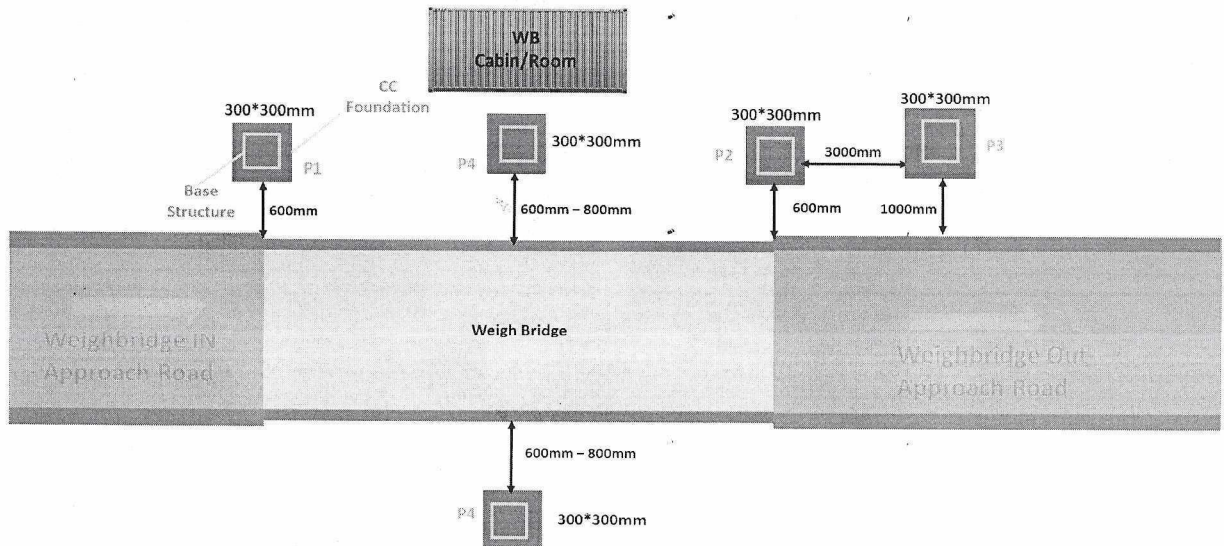
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3. Wiring & Cabling Architecture of Weighbridges with Mandatory Components



4. Civil Construction at Weighbridges



Note: The layout of civil structure may be changed by ME/AME as per site feasibility

5. Base Structure Guidelines for Equipment Installation:

- (1) Position sensors should be mounted at height of 1500mm from platform of the weighbridge.
- (2) Height of Poles from platform of the weighbridge should be,-
 - (i) P1 Pole: 1800-2000mm
 - (ii) P2 Pole: 1800-2000mm
 - (iii) P3 Pole: 2800-3000mm

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(iv) P4 Pole: 7800-8000mm

Note: L-shape or U-shape Pole may be used as per site conditions.

(3) Cameras should be mounted at a height of,-

(i) C1 Camera: 7600-7800mm, all four corners of WB platform must be visible.

(ii) C2 Camera: 1800-2000mm

(iii) C3 Camera: Depending upon on the height of room/cabin covering digitizer and PC screen.

Note: IP bullet camera installation guide enclosed at Annexure-2.

(4) For pit type WB: Side railing should be erected on both sides of WB having at least 600mm height.

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Annexure.-1

Technical Specification

1. RFID Reader

- **Frequency & Compliance**
Operating Frequency: 865-867 MHz (as per WPC regulations in India)
RF Protocol: EPC global Gen2 (ISO 18000-6C)
Antenna Type: Inbuilt or additional
RF Power Output: Adjustable (up to 1W or 30 dBm, as per WPC limits)
- **Read Performance**
Read Range: 3-6 meters
Reading Speed: Up to 200 tags per second
Tag Type: Passive UHF RFID tags (Gen2 UHF)
Multi-tag Reading: Supports anti-collision for multiple tag reading
- **Connectivity & Interfaces**
Communication: Socket Communication (Mandatory)
Interface: Ethernet
Power Supply adapter: 9V to 24V DC input
- **Environmental & Mechanical Features**
Operating Temperature: -20°C to +60°C
Protection Rating: IP65 or higher (for outdoor use)
Mounting: Pole-mount or wall-mount options
Housing Material: Rugged, weatherproof enclosure

2. Position Sensors

S.N.	Parameter	Technical Specifications
1	Type	Ultrasonic Sensor
2	Sensing range	200-2000 mm
3	Adjustment range	240-2000 mm
4	Transducer Frequency	Approx. 85 kHz or better
5	Opening Voltage	12 V DC, ripple 10%ss
6	No-load supply current	<50 mA
7	Degree of protection	IP 65
8	Output type	2 switch outputs PNP, NO/NC, programmable
9	Note: The sensor should have configurable sonic beam with sensing range and programmable blasting time.	

3. IP Bullet Camera – C1 and C3

S.N.	Parameter	Technical Specifications
1	Image sensor	1/2.8" Progressive scan CMOS sensor
2	Resolution	Minimum 2 Mega Pixel; 1920X1080 @ 25/30 FPS or better
4	IR Range	IR 30-meter or higher
5	Communication Interface	Ethernet RJ45 10M/100M
6	Power requirement	PoE+/ 12VDC





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	IEC62471:2006
8	Protection Function: Transient over peak suppression or use external surge level
9	Housing Material: Die-casting aluminium alloy
10	Power Input: As per OEM Standard
11	Eye safety certificate / Equivalent test report to be submitted

5. Controller

S.N.	Parameter	Technical Specifications
1	Input / Output Channels	• 8-channel isolated digital input/output module. • Communication: MODBUS (Ethernet/RTU) • Output Relay Contact Capacity : 10A each @ 240VAC, 10A @ 28VDC . • Input Isolation : Optically isolated • DI Register Address : 0,1,2,3,4,5,6,7 • DO Register Address : 0,1,2,3,4,5,6,7 • Power: 12V DC • LED indicators per I/O channel • DIN rail mountable • Operating temp: -20°C to +70°C • ESD & surge protection
2	Power Supply (Module)	• Input: 100–240V AC, 50/60Hz • Output: 12V DC, 10A (min) • Overload, over voltage, short-circuit protection • Efficiency $\geq 80\%$ • Enclosed, industrial-grade • CE/ISO certified • DIN rail mountable(optional)
3	Enclosure	Type: Wall-mounted / Pole-mounted / Outdoor Application: Housing of above mentioned items Standard Compliance: IEC 60529 (IP Rating) or similar Protection Class: IP 65 & Outdoor support Doors: Hinged front door with concealed hinges and gasket sealing Cable Entry: Bottom or top gland plate, detachable, 2.0 mm thickness Body Material: Cold Rolled Close Annealed (CRCA) Sheet Steel, 2.0 mm for body Ventilation: Louvers with dust filters / optional cooling fan for heat dissipation Electrical & Earthing: Provision must be there Mounting and Accessories: Wall mounting brackets / Floor stand (with base frame) Relative Humidity: up to 95% non-condensing Installation Location: Indoor / Outdoor / Coastal / Industrial area

6. Pole & Side Railing

S.N.	Parameter	Technical Specifications
1	Pole Details	Pole shall be hot dip galvanize
2	Civil Construction details	The pole shall be installed on a precast or cast in CC foundation with nuts & washers and with a set of four foundation bolt for greater rigidity basis the soil bearing capacity of the actual site.
3	Pole Size	As mentioned in Point 5 of SoP
4	Side Railing	For Pit-Type WB, it should be created on both sides of WB having atleast 600mm height.

7. Personal Computer

S.N.	Parameter	Technical Specifications
1	Architecture	Intel i5 processor 10 th Generation or higher
2	Form Factor	Tower or Rack
3	Processor	1x 6 Core or Above
4	Memory	16 GB DIMM Memory or More
5	Expansion Slots	Up to 4 x PCIe 3.0
6	HDD/SSD	1 TB GB (minimum)
7	Operating System	Windows 11
8	Size	Full HD LED Monitor or higher
9	Connectivity	1VGA Port,1 HDMI Port

8. PoE Network Switch

S.N.	Parameter	Technical Specifications
1	Port Density	08-port 10/100/1000TX switch
2	PoE	PoE ports with Minimum 190W total power budget
3	VLAN	Support VLAN feature
4	L2 Protocol	Support L2 Protocol
5	Management Interface	CLI/GUI

9. Traffic Lights

S.N.	Parameter	Technical Specifications
1	Description-Two colour (Red and Green)	Stop/Go -Red and Green Traffic Light
2	LED LIFE	1,00,000 HRS
3	Type	SMD 3528 or Better
4	Housing	200MM/300MM-Standard
5	LED	High bright
6	Input Voltage	12V DC

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10. Boom Barrier with Safety Sensor

S.N.	Parameter	Technical Specifications
1	Type	Outdoor
2	Ingress Protection	IP 54 or Higher
3	Interfaces	Wired (Open, Close, COM)
4	Max. Boom Pole Length	3 Mtr. Barrier
5	Opening & Closing Time	≤ 5.0 seconds
6	Safety Sensor	Mandate to installed (As per requirement)

11. LED Display

SI can supply and install either Type-1 or Type-2 LED Display.

11.1 Protocol: MQTT (Type – 1)

S.N.	Parameter	Technical Specifications
1	Type	P10 Red Color LED
2	LED Intensity	5500mCD or higher
3	Visibility Range	Greater than 20 Meter
4	Refresh Rate	50Hz-100Hz
5	Display Support	True type fonts and adjustable based on the Operating system requirement
6	Size	36" X 12" or above
7	Communication	Ethernet
8	MQTT Configuration	Display Configuration 1. Set the hostname as the computer IP address. 2. Set the port number to 1883. 3. Set the subscribe topic to "iot_display/application_to_display". 4. Set the publish topic to "iot_display/display_to_application". After receiving data on the topic "iot_display/application_to_display", display the received data on the screen, and also publish the same received data on the topic "iot_display/display_to_application".
9	Language	English & Numeric
10	Rating (For Outdoor Use)	IP 56 or better

11.2 Protocol: API (Type – 2)

SI#	Parameter	Technical Specifications
1	Type	P10 Red Color LED
2	LED Intensity	5500mCD or higher
3	Visibility Range	Greater than 20 Meter
4	Refresh Rate	50Hz-100Hz
5	Display Support	True type fonts and adjustable based on the

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		Operating system requirement
6	Size	36" X 12" or above
7	Communication	Ethernet
8	API Configuration	URL: "/display_data" Method :POST Payload: {"data": " " } Response : {"status": 200 }
9	Language	English & Numeric
10	Rating (For Outdoor Use)	IP 56 or better

12. PA System

Sl#	Parameter	Technical Specifications
1	Power Output	37W Max., 30W RMS at 10% THD, 27W RMS at 5% THD
2	Output Regulation	≤ 2 dB, no load to full load at 1kHz
3	Input Channels	2 × Mic. 0.5mV/2.5kΩ, 1 × Aux 50mV/330kΩ
4	Speaker Outputs	4Ω, 8Ω, 16Ω & 100V
1	Input Power	30W RMS/45W Max.
2	Frequency Response	240-10,000Hz

13. Internet Connectivity

Dedicated 20 Mbps Internet Leased Line (ILL) / Broadband / Airfiber connection with

a) Public Static IP

b) NAT and Port Forwarding support through Service Provider-supported router

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Annexure -2

IP bullet camera – C2 installation guide used for number plate reading

1. Recommended Installation Location

1.1 Camera Placement

- Install the camera facing incoming vehicles.
- Camera must be installed after position sensor.
- Preferred installation: Straight vehicle approach

2.2 Avoid Installing Camera:

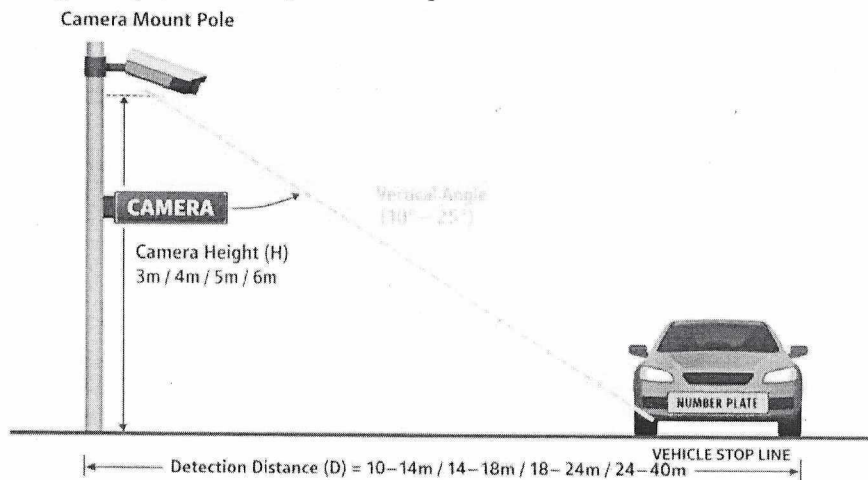
- On curved roads
- At sharp turn entries
- Where headlights directly face the lens at close range
- Where vehicles stop too close to the camera (< 6 meters)

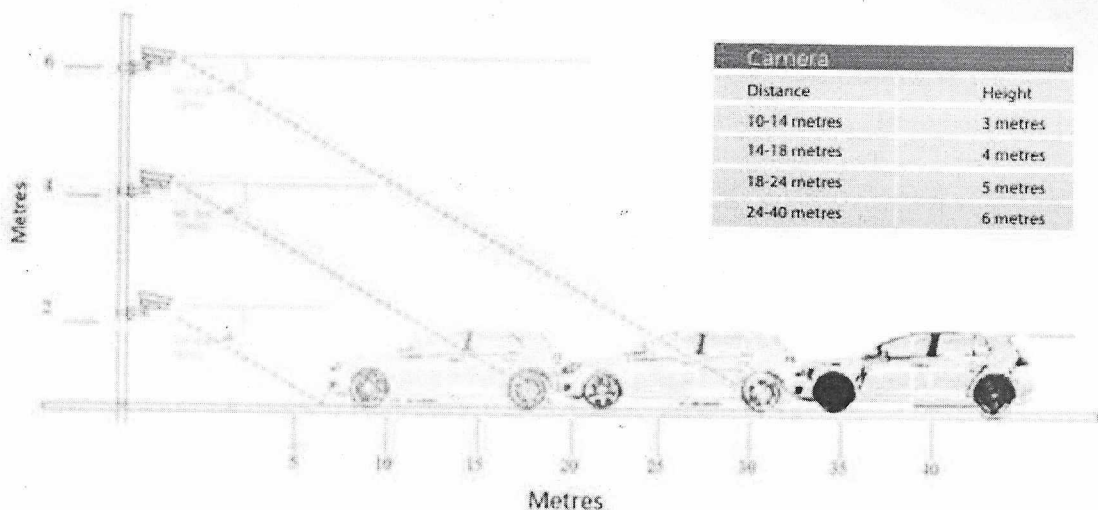
3. Distance vs Camera Height Guideline

Vehicle Detection Distance	Recommended Camera Height
10 – 14 meters	3 meters
14 – 18 meters	4 meters
18 – 24 meters	5 meters
24 – 40 meters	6 meters

Important Notes:

- Height is measured from ground level to camera lens center
- Do not exceed 6 meters height
- Lower height improves clarity for small plates

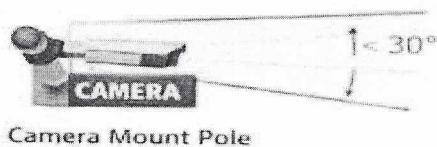




4. Camera Angle Guidelines

4.1 Horizontal Angle (Side Angle)

- Camera should face the number plate as straight as possible
- Maximum allowed horizontal angle: ≤ 30 degrees
- Recommended angle: 15–20 degrees



Camera Mount Pole

- Recommended Angle
- ✓ $\leq 20^\circ$ angle
- ✗ Maximum Allowed Angle $< 30^\circ$ angle

4.2 Vertical Angle (Tilt Angle)

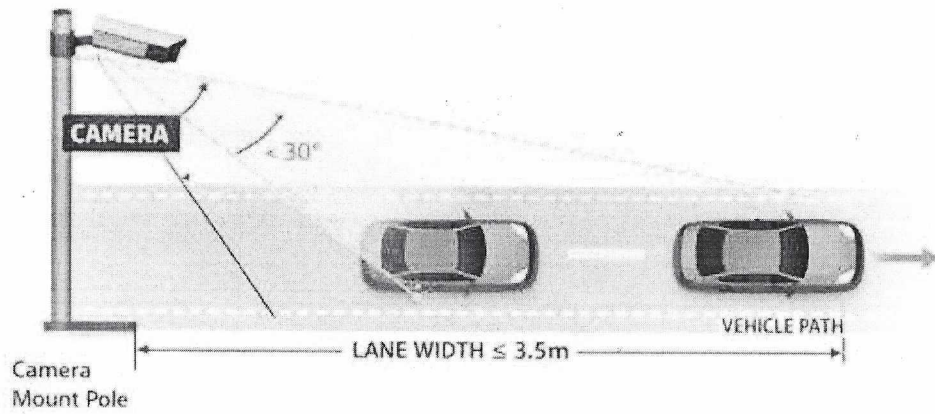
- Camera tilt (downward): ≤ 25 degrees
- Ideal tilt range: 10–15 degrees
- Excessive tilt reduces OCR accuracy

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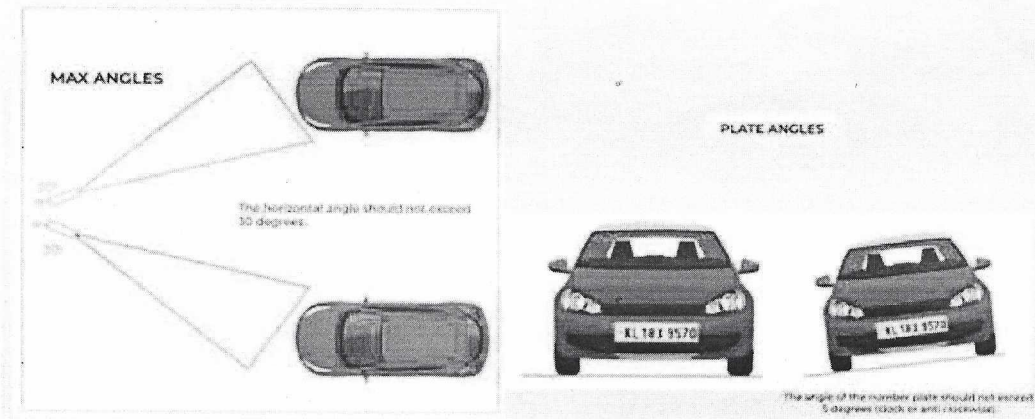
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Camera Mount Pole



- ✓ Recommended Angle $\leq 20^\circ$ angle
- ✗ Maximum Allowed Angle $< 30^\circ$ angle

Camera Angel



5. Lane & Field of View Guidelines

- One camera = one lane
- Lane width should ideally be ≤ 3.5 meters
- Number plate should appear:
- Fully visible
- Centered in frame

6. Lighting & Environment Conditions

6.1 Lighting

- Avoid strong backlight or direct sunlight into lens
- Ensure IR reflection is not blocked by:
- Glass
- Plastic covers

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- Do not install near reflective surfaces

6.2 Night & Low Light

- Ensure IR range is sufficient for target distance
- Avoid installing camera too close to:
- White walls
- Metal boards (causes IR reflection)

7. Mounting & Stability Requirements

- Use rigid metal pole or wall mount
- No vibration or shaking
- Camera must not move due to wind or vehicle movement
- Tighten all mounting screws firmly





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